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# Smart patch for swallowing guidance

## A Non-Invasive, Closed-Loop Smart Patch for Safe Swallowing Guidance in Stroke Survivors: Technologies, Mechanisms, Control Systems, and Future Directions

### Quick Reference

### Key Findings Table

| Theme | Summary | Supporting Citations |
|---|---|---|
| Multimodal Sensing Integration | Combining EMG, MMG, piezoelectric, acoustic, accelerometric, and optical sensors improves detection, but standardization and sensor fusion remain challenges. | 1 2 3 |
| Closed-Loop Control & AI Optimization | Closed-loop neuromodulation with machine learning enables real-time, personalized feedback; robust adaptive AI models are needed for clinical use. | 4 5 6 |
| Safety & User-Centered Design | Mild adverse events (skin irritation) are common; co-design and personalized interfaces improve adoption and adherence. | 7 8 9 |
| Gaps in Clinical Validation & Standardization | Lack of large-scale validation, especially in diverse populations; standardized calibration/testing protocols are essential. | 10 11 12 |
| Multidisciplinary & Translational Approaches | Collaboration across neuroscience, materials science, engineering, and clinical rehab is critical for translation. | 13 14 15 |

### Direct Answer

Current smart patch and wearable technologies show promise in monitoring and guiding swallowing events in stroke survivors—using integrated multimodal sensors and closed-loop control systems paired with AI. However, further clinical validation, improved sensor fusion, and standardized calibration protocols are needed. Future developments should focus on comprehensive sensing (EMG, acoustic, accelerometric, optical), advanced machine learning for real-time personalized feedback, and user-centered design to enhance safety, adherence, and clinical efficacy in diverse patient populations.

### Study Scope

- **Time Period:** Recent advances (last 5–10 years), with emphasis on current and emerging technologies.
- **Disciplines:** Neuroscience, materials science, biomedical engineering, clinical rehabilitation, digital health, regulatory science.

- **Methods:** Meta-analysis, systematic reviews, original research, clinical trials, device development, user-centered design studies.

### Assumptions & Limitations

- Most findings are based on hospital-based studies; community/home-care validation is limited.
- Device efficacy and safety data are often short-term; long-term outcomes and adherence require further study.
- Underrepresentation of women and minorities in clinical trials may limit generalizability.
- Sensor fusion and AI models are promising but not yet fully robust for real-world deployment.
- Regulatory and standardization frameworks are evolving and may differ by region.

### Suggested Further Research

- Large-scale, multicenter clinical trials with standardized protocols.
- Integration of multimodal sensors with advanced AI for robust, real-time swallowing event detection.
- Inclusive, user-centered design and validation in diverse populations.
- Regulatory harmonization and post-market surveillance for wearable swallowing devices.
- Multidisciplinary collaboration for translational research and clinical adoption.

## 1. Introduction

### Background and Rationale

Dysphagia, or difficulty swallowing, is a prevalent and serious complication in stroke survivors, affecting up to 50% of patients and significantly increasing the risk of aspiration, pneumonia, and fatal choking. Conventional interventions—ranging from dietary modifications to therapist-led exercises—often lack real-time feedback and objective monitoring, leading to inconsistent outcomes and limited prevention of life-threatening events. The emergence of non-invasive, closed-loop smart patch technologies offers a transformative approach: real-time, personalized guidance of throat movements during swallowing, leveraging advanced sensors, machine learning, and adaptive control systems to prevent choking and improve rehabilitation outcomes. However, gaps remain in clinical validation, sensor integration, and user-centered design, necessitating a comprehensive review of current technologies, physiological mechanisms, control systems, and future directions for safe and effective deployment in stroke rehabilitation [13] [15] [16] [17].

## 2. Smart Patch and Wearable Technologies for Swallowing Assistance

### Current Technologies and Clinical Effectiveness

- **Throat-Wearable Sensing Systems (TWSS):** Flexible patches with multimodal sensors (EMG, accelerometry, acoustics) achieve high accuracy (>90%) in swallowing event detection and laryngeal activity assessment, outperforming conventional models and supporting personalized healthcare [18].

- **Multimodal Aspiration Prevention Systems (MAPS):** Wearable-based interventions with risk warning and multidisciplinary collaboration significantly reduce aspiration and pneumonia, improve psychological well-being, and patient satisfaction, though swallowing function improvement may not always be statistically significant [15].
- **Neuromuscular Electrical Stimulation (NMES):** Combined with traditional therapy, NMES improves swallowing function, electromyography scores, and transit times, serving as an effective wearable adjunct [19].
- **Acoustic and Accelerometric Devices:** Offer accessible, home-based assessment but variable accuracy; only half of studies report >90% accuracy, highlighting the need for standardization [20].
- **Digital Health Tools:** mHealth apps, telepractice, and game-based biofeedback provide personalized exercises and real-time feedback, but digital unfamiliarity and technical issues limit adoption [21].
- **User Perspectives:** Stroke survivors and therapists value user-friendly, flexible apps and wearable devices that provide feedback and strengthen patient-therapist relationships [16].
- **Clinical Translation Gap:** Despite advances, standardized evaluation and optimization for swallowing function and choking prevention are needed [22] [23].

#### Synthesis

Wearable and smart patch technologies are increasingly effective in monitoring and guiding swallowing, but their clinical translation is hampered by variability in accuracy, lack of standardization, and limited user adoption. Multimodal sensing and digital feedback are promising, yet require further validation and user-centered design [15] [16] [18] [19] [20] [21] [22] [23].

### Flexible Sensor Materials and Designs

- **Fiber Optic & Polymer Sensors:** Hetero-core fiber optic stretch and pressure sensors, rubrene single crystal strain sensors, and flexible polymers offer high sensitivity and stability for non-invasive swallowing assessment [24] [25] [26].
- **Graphene-Based Electronic Skin:** Provides conformal attachment, repeatable resistance changes, and low drift for monitoring swallowing and respiration [27].
- **Hydrogel-Based Sensors:** Cross-linked hydrogels (e.g., l-glutamic acid, MXene composites) exhibit high stretchability, toughness, self-recovery, and sensitivity, suitable for detecting subtle throat motions [28] [29].
- **Nanomaterial Integration:** MXene, carbon nanomaterials, and liquid metal composites enhance mechanical properties, conductivity, and environmental adaptability [30] [31] [32] [33].
- **Optical Trackers & MMG Sensors:** Near-infrared sensing and mechanomyography enable long-term laryngeal activity monitoring and analysis of swallowing-respiratory coordination [34] [35].

#### Synthesis

Material innovations—especially hydrogel, graphene, and nanomaterial composites—are driving advances in sensitivity, durability, and biocompatibility of throat-wearable sensors. Multifunctional designs enable robust detection of swallowing and related motions, but long-term stability and integration remain challenges [24] [25] [27] [28] [29] [30] [31] [32] [33].

### Multimodal Sensing and Machine Learning Integration

- **Sensor Fusion:** Combining EMG, MMG, piezoelectric, acoustic, and optical sensors improves swallowing event detection and respiratory-swallow coordination monitoring [35] [36] [37] [38] [39].
- **Machine Learning Models:** Deep learning (CNN, LSTM), few-shot learning, and transfer learning enhance classification accuracy (>80–92%) and robustness to noise [38] [39] [40].
- **Challenges:** Sensor cross-talk, signal heterogeneity, and calibration standardization are recurring issues [1] [2] [3].

### Synthesis

Multimodal sensing and machine learning integration are essential for accurate, real-time swallowing event detection. Advanced AI models improve robustness, but sensor fusion and calibration protocols need refinement for clinical reliability [1] [2] [3] [35] [36] [37] [38] [39].

### Limitations and Standardization Challenges

- **Noise & Artifacts:** Acoustic devices are susceptible to speech and external noise, leading to high false positives and moderate precision (~42–50%) [41].
- **Sensor Placement & Variability:** Single-point sensors assume signal symmetry, but evidence suggests asymmetry and intra-personal variability; multi-sensor arrays are needed [1].
- **Small Sample Sizes & Protocol Heterogeneity:** Many studies have imbalanced class distributions and lack standardized assessment protocols [20].
- **Machine Learning Limitations:** Precision and false positives remain concerns; larger, balanced datasets and standardized recording conditions are needed [18] [42].

### Synthesis

Current devices face limitations in accuracy, standardization, and clinical translation due to noise, sensor variability, and protocol heterogeneity. Improved sensor arrays, signal processing, and standardized protocols are critical for reliable swallowing detection [1] [20] [41] [42].

## 3. Physiological Mechanisms and Monitoring of Throat Movements

### Neurophysiology of Swallowing in Stroke Survivors

- **Neural Circuits:** Swallowing is controlled by brainstem central pattern generators (NTS, nucleus ambiguus) and higher cortical regions, integrating reflexive and voluntary phases [43] [44] [45] [46] [47] [48].

- **Cortical Reorganization:** Unilateral stroke impairs pharyngeal phase initiation; recovery involves increased representation in the unaffected hemisphere and neuroplasticity [44] [46] [49].

- **Biomarkers:** Motor evoked potentials (MEPs), BDNF levels, and functional connectivity changes indicate recovery progression [48] [50].

#### Synthesis

Swallowing involves complex sensorimotor integration, with stroke disrupting both brainstem and cortical control. Recovery is marked by neuroplasticity and biomarker changes, guiding rehabilitation strategies [43] [44] [45] [46] [47] [48].

### Sensorimotor Feedback and Airway Protection

- **Pharyngeal & Laryngeal Afferents:** Sensory inputs from glossopharyngeal and superior laryngeal nerves trigger and modulate swallowing reflexes, essential for airway protection [51] [52] [53] [54].

- **Aspiration Risk:** Reduced sensitivity or afferent dysfunction increases aspiration risk; interventions targeting sensory pathways (e.g., pharyngeal electrical stimulation) aim to restore function [52] [55].

#### Synthesis

Sensorimotor feedback from pharyngeal and laryngeal afferents is critical for safe swallowing and aspiration prevention. Stroke-induced dysfunction necessitates targeted rehabilitation to restore sensory input [51] [52] [53] [54].

### Real-Time Monitoring Techniques

- **Ultrasound:** Simple, repeatable, real-time imaging of tongue, hyoid, and larynx motion; excels in dynamic soft tissue visualization [56] [57] [58].

- **Real-Time MRI:** High temporal resolution, robust quantitative assessment, superior soft tissue contrast; less accessible than ultrasound [59] [60] [61] [62].

- **Respiratory Inductance Plethysmography (RIP):** Non-invasive, automated swallow detection with accuracy up to 86%; machine learning enhances performance [63] [64] [65].

- **Multimodal Wearable Sensors:** Combine laryngeal movement, submental EMG, and nasal airflow for comprehensive assessment [66].

#### Synthesis

Dynamic, non-invasive monitoring techniques—ultrasound, MRI, RIP, and multimodal sensors—enable real-time assessment of swallowing and respiratory coordination, supporting objective evaluation and feedback in rehabilitation [56] [59] [63] [66].

## 4. Closed-Loop Control Systems in Swallowing Rehabilitation

### Design and Implementation of Closed-Loop Systems

- **Architecture:** Integration of sensors, controllers, actuators, and safety frameworks; real-time monitoring and automatic adjustment based on physiological feedback [4] [67] [68] [69].
- **Safety Governance:** Layered frameworks separate performance optimization from safety enforcement; FDA guidance and risk management protocols standardize development [4] [69].

#### Synthesis

Closed-loop systems require robust integration of sensing, control, and actuation, with layered safety governance to ensure patient safety and regulatory compliance [4] [67] [68] [69].

### Control Algorithms and Neuromodulation Strategies

- **Neuromodulation Techniques:** NMES, rTMS, taVNS, and MI-BCI-controlled FES are effective for guiding/correcting swallowing actions [4] [70] [71] [72] [73].
- **Adaptive Algorithms:** PID, LQG, and AI-driven controllers personalize stimulation based on real-time biomarkers [4] [70] [74].
- **MI-BCI Decoding:** EEG-based detection of swallowing intention enables alignment of stimulation with voluntary effort, improving safety and efficacy [70] [75] [76].

#### Synthesis

Effective closed-loop neuromodulation combines advanced control algorithms with real-time biomarker feedback, enabling personalized, adaptive rehabilitation for swallowing dysfunction [4] [70] [71] [72] [73] [75] [76].

### Clinical Efficacy and Safety of Closed-Loop Interventions

- **MI-BCI-FES Systems:** Demonstrate technical feasibility, acceptable usability, and safety in clinical trials; classification accuracy ~71.5%, adherence 100% [70] [77] [78] [79].
- **taVNS:** Combined with conventional therapy, improves swallowing function, increases BDNF, decreases inflammatory markers, and is well-tolerated [72] [80].
- **Combined Therapies:** NMES with adjuncts (ice stimulation, cervical mobilization) yields superior outcomes and safety [81] [82].

#### Synthesis

Closed-loop interventions, especially MI-BCI-controlled FES and taVNS, are clinically effective and safe, with combined therapies further enhancing outcomes. Standardized protocols and safety frameworks are essential for broader adoption [70] [72] [77] [78] [79] [80] [81] [82].

### Biomarker Feedback and Adaptive Control

- **Biomarker Selection:** Neural, neurochemical, and physiological signals guide real-time feedback and adaptive control [70] [74] [83] [84].
- **AI-Driven Control:** Data-driven approaches enhance biomarker detection and control policy optimization, improving therapy effectiveness [6] [85].
- **Artifact Mitigation:** State-aware feedback and adaptive signal processing ensure high-fidelity neural recording during stimulation [86] [87].

#### Synthesis

Adaptive control strategies leveraging real-time biomarker feedback and advanced AI models optimize closed-loop neuromodulation, enabling personalized, responsive swallowing rehabilitation [6] [70] [74] [83] [84] [85] [86] [87].

### 5. Safety and Efficacy Considerations in Stroke Rehabilitation Devices

#### Safety Profiles and Adverse Events

- **Mild Adverse Events:** Skin irritation (abrasions, blisters, dermatitis) is common but manageable; device design improvements reduce incidence [7] [88].
- **NMES & Digital Tools:** Generally safe and well-tolerated; adherence data are limited [89] [90].
- **Management Strategies:** Iterative design, user training, and workflow integration mitigate adverse events and improve acceptance [7] [91].

#### Synthesis

Wearable swallowing devices are safe, with mostly mild adverse events. Continuous design improvement and user training are key to minimizing risks and enhancing tolerability [7] [88] [89] [90] [91].

#### Efficacy of Combined and Adjunct Therapies

- **NMES with Adjuncts:** Combined with ice stimulation, cervical mobilization, or effortful swallowing, NMES yields superior functional recovery and safety [81] [82] [92].
- **Non-Invasive Brain Stimulation:** rTMS and tDCS are effective and safe, with rTMS showing larger effect sizes; optimal protocols remain unclear [92] [93] [94] [95] [96] [97].
- **Digital Rehabilitation:** Personalized, remote, and game-based interventions improve outcomes and quality of life [98] [99].

#### Synthesis

Combined therapies and non-invasive brain stimulation enhance swallowing recovery and safety. Digital tools offer additional benefits, but protocol optimization and user engagement are needed for sustained efficacy [81] [82] [92] [93] [94] [95] [96] [97] [98] [99].

### User-Centered Design and Adoption

- **Co-Design & Personalization:** Workshops, video prototypes, and community engagement clarify needs and improve usability [8] [100].
- **Digital Familiarity & Access:** Barriers include unfamiliarity, access issues, and technical challenges; simple, evidence-based interfaces are preferred [21] [101].
- **Stakeholder Involvement:** Collaboration among survivors, caregivers, clinicians, and industry is essential for adoption [102] [103].

#### Synthesis

User-centered design—through co-design, personalization, and stakeholder engagement—improves adoption and adherence to digital swallowing rehabilitation technologies. Addressing digital barriers and usability is critical for widespread acceptance [8] [21] [100] [101] [102] [103].

### Multidisciplinary Protocols and Clinical Integration

- **Multimodal Systems:** Risk warning, standardized procedures, and multidisciplinary collaboration reduce aspiration and pneumonia, improve well-being and satisfaction [15].
- **Combined Rehabilitation:** NMES, swallowing exercises, and neuromuscular taping are effective; tailored, progressive programs optimize outcomes [104] [105].
- **Instrumental Assessments:** VFSS, FEES, and mHealth-based programs support safe, evidence-based deployment [106] [107].

#### Synthesis

Multidisciplinary care, standardized protocols, and clinical workflow integration are vital for safe and effective deployment of swallowing rehabilitation devices. Evidence-based, tailored approaches maximize safety and efficacy [15] [104] [105] [106] [107].

## 6. Gaps and Future Research Directions in Smart Patch Development for Dysphagia

### Current Gaps and Limitations

- **Sensor Integration:** Limited monitoring of respiration-swallow coordination; lack of closed-loop systems with real-time personalized feedback [10] [13] [108] [109].
- **Clinical Validation:** Insufficient evidence in diverse populations; most studies are hospital-based, with limited community/home-care testing [10] [11] [12].
- **Standardization:** Need for calibration/testing protocols and comprehensive, annotated datasets [2] [110].

#### Synthesis

Smart patch technologies face gaps in sensor integration, closed-loop feedback, and clinical validation. Standardized protocols and inclusive trials are needed for effective translation [2] [10] [11] [12] [13] [108] [109] [110].

### Multimodal Sensing and AI Optimization

- **Sensor Combinations:** EMG, piezoelectric, optical, and MMG sensors capture mastication and swallowing dynamics; multimodal integration enhances accuracy [111] [112].
- **AI Algorithms:** Neural networks, deep learning, and edge AI processing improve real-time swallowing identification and personalized feedback [6] [110] [113].
- **Challenges:** Sensor stability, data security, biocompatibility, and computational efficiency remain barriers [114].

### Synthesis

Multimodal sensing and advanced AI models are key to real-time, personalized swallowing guidance. Material and computational innovations are needed to overcome current limitations [6] [110] [111] [112] [113] [114].

### Clinical Validation and Trial Design

- **Trial Designs:** Double-blind, randomized controlled trials with sham controls and multimodal neurostimulation arms are recommended [9] [115] [116] [117].
- **Inclusion & Diversity:** Tailored inclusion criteria and remote assessment strategies improve recruitment and retention of diverse populations [9] [118] [119].
- **Long-Term Outcomes:** Larger sample sizes, standardized protocols, and objective assessments are needed for definitive efficacy and safety evaluation [12] [120].

### Synthesis

Rigorous, inclusive clinical trials with standardized protocols are essential for validating smart patch efficacy and safety in diverse populations. Long-term outcome data and objective assessments will drive regulatory approval and adoption [9] [12] [115] [116] [117] [118] [119] [120].

### Regulatory and Adoption Challenges

- **Calibration & Testing:** Physiologically grounded calibration (e.g., voice segment durations) improves clinical interpretability [3].
- **Regulatory Pathways:** Premarket Approval (PMA) with robust safety/efficacy data is preferred; postmarket surveillance is critical [11] [121].
- **Global Adoption:** Barriers include limited validation, heterogeneity in protocols, and infrastructure limitations; multidisciplinary teams and patient-focused design are needed [122] [123] [124].

### Synthesis

Regulatory harmonization, standardized calibration, and patient-focused design are necessary for global adoption of smart patch technologies. Multidisciplinary collaboration and ongoing surveillance ensure safety and effectiveness [3] [11] [121] [122] [123] [124].

### 7. Conclusion

#### Summary and Outlook

Non-invasive, closed-loop smart patches represent a transformative advance in dysphagia management for stroke survivors, offering real-time, personalized guidance to prevent fatal choking. Key themes include multimodal sensor integration, adaptive closed-loop control systems enhanced by AI, and the critical importance of safety and user-centered design. However, challenges in clinical validation, standardization, and sensor fusion persist, underscoring the need for multidisciplinary collaborations and rigorous trials. Future research should focus on inclusive, large-scale validation, advanced AI-driven feedback, and regulatory harmonization to translate these innovations into effective clinical solutions [13] [18] [73] [88].

**Creative Insight:**
A promising avenue is the development of ultra-compliant, bio-integrated hydrogel smart patches with embedded edge AI, combining flexible piezoelectric and graphene-based sensors with multimodal data streams (EMG, acoustics, respiratory signals) and real-time closed-loop control algorithms. This would enable dynamic, personalized rehabilitation protocols that adapt to patient needs and environmental conditions.

**Synthesis Statement:**
In summary, the research reveals strong potential for non-invasive, closed-loop smart patches in preventing fatal choking among stroke survivors by guiding swallowing movements with high accuracy. Integration of multimodal sensors, adaptive closed-loop control, and user-centered design are essential, but clinical validation, standardization, and multidisciplinary collaboration remain critical for effective translation into practice.

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